- *Google Cloud Monitoring (https://cloud.google.com/monitoring)*—Part of the Google Cloud's operations suite; it enables you to collect metrics, events, and metadata from Google Cloud, AWS, hosted uptime probes, and application instrumentation.
- *Google Cloud Managed Service for Prometheus (http://mng.bz/GyaM)*—Runs Prometheus-based collectors as DaemonSets in Kubernetes and ensures scalability by only scraping targets on colocated nodes and pushing the scraped metrics to the central data store (Monarch). You can use PromQL to query the metrics, for alerts or, for example, as a data source in Grafana.

Figure 5.3 shows an example metrics explorer output—here, the Metrics Explorer being part of the Azure Monitor Metrics service. Next, let's look at open source offerings for metrics LTS and federation.

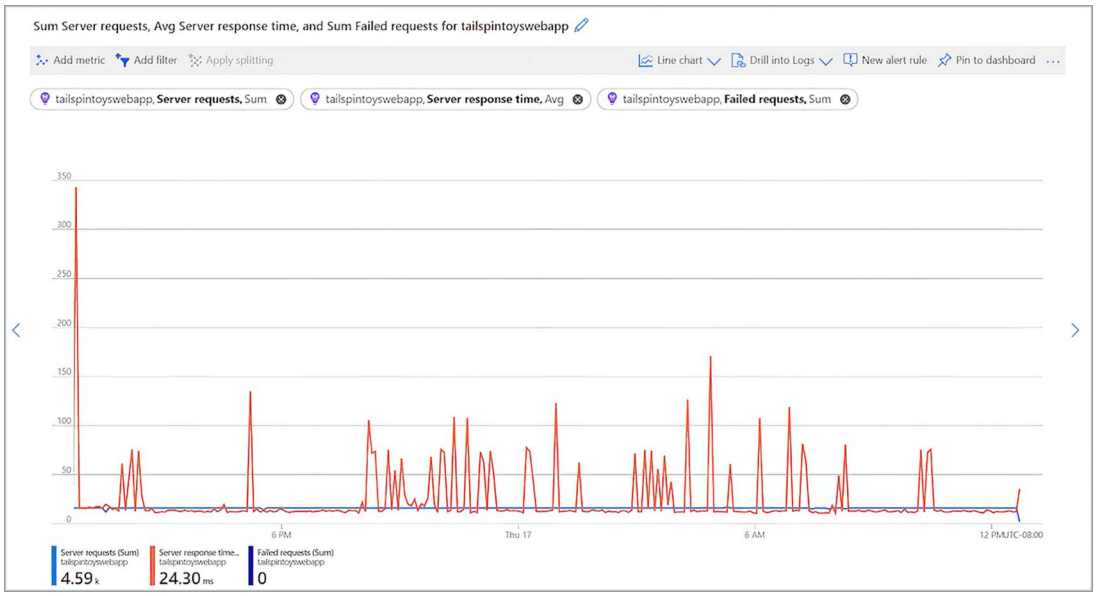


Figure 5.3 Azure Metrics Explorer example output (Azure docs, http://mng.bz/Y1Bo)

5.3.2 Open source metrics backends

The metrics backend space is rather crowded, which is good for you, in terms of options available, but it can also be challenging to keep track of developments and make the best decision for workload:

- *CNCF Cortex and Thanos*—Two CNCF projects with different philosophies about how to ingest and query Prometheus timeseries. Cortex (https://cortexmetrics.io/) is used in Amazon Managed Service for Prometheus; and Thanos (https://thanos.io/), founded by Improbable, is used in a variety of cases by many large